

Temporal Coordination: Turns, Overlap, and Persistence

Temporal Coordination: Turns, Overlap, and Persistence I

The original PPPiP form is synchronous: both partners mark the surface at the same time. DigiPPPiP separates synchrony from relationality. Synchronous sessions still matter because joint action requires partners to coordinate bodies, intentions, and timing, and because active interaction is a strong candidate context for inter-brain synchrony in cooperative and naturalistic tasks [sebanz2006joint; czeszumski2020hyperscanning; czeszumski2022cooperative]. Emerging work on online group interaction suggests that neural alignment can also appear under virtual conditions when participants remain actively engaged [azhari2025online]. Yet synchrony is not the only temporal form that can support intimacy, play, or shared meaning.

Temporal Coordination: Turns, Overlap, and Persistence II

Remote and delayed modes should not be described as if distance has been solved. Distance changes the cost of coordination, common ground, informal repair, and collaboration readiness [Clark1991]. Grounding theory adds a finer vocabulary: partners need evidence that a mark, gesture, or prompt has been perceived and understood well enough for the next contribution [Clark1991]. Recent video-mediated collaborative-drawing work makes that requirement concrete: proposal sequences depend on verbal, embodied, and screen-based actions, and the person controlling the drawing tool can become pivotal in coordinating joint decisions [Oittinen2005]. DigiPPPiP's temporal design should therefore measure not only whether users were online together, but how the canvas supported acknowledgement, repair, turn completion, tool-control transfer, and later re-entry into the same shared artifact.

Temporal Coordination: Turns, Overlap, and Persistence III

Semisynchronous DigiPPPiP introduces alternation. Partners are present in the same session but take turns or contribute in bursts. That mode preserves mutual responsiveness while creating short prediction intervals: each mark becomes a cue for anticipating the partner's next action. It also makes repair visible: an undo, hesitation, or spoken clarification can become part of the event stream rather than an invisible failure. This is a natural bridge between the modeling lens in [[@sec:active_inference](#)] and the formal narrative account in [[@sec:formalisms_appendix](#)].

Temporal Coordination: Turns, Overlap, and Persistence IV

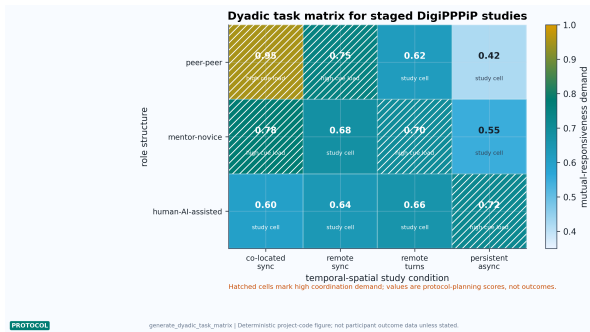
Asynchronous DigiPPPiP changes the practice more radically. Partners contribute at separate times, making the canvas a persistent relational object. Asynchronous mental-health technologies show that clinical or supportive exchange need not always be real-time to be meaningful [lagera2023asynchronous]. For DigiPPPiP, absence becomes part of the form: waiting, revisiting, and responding create a slower temporal arc of curiosity and resolution. Digital-intimacy research is relevant here because many strong-tie technologies rely on awareness, traces, rituals, and low-pressure presence rather than continuous conversation [hasse nzahl2012love;vetere2005mediating].

Temporal Coordination: Turns, Overlap, and Persistence V

The temporal primitives are overlap, order, delay, persistence, and repair. Overlap supports co-action; order supports anticipation; delay supports reflection; persistence supports return; repair supports trust after confusion. These primitives are more fundamental than labels such as live, remote, or asynchronous because the same interface can instantiate them differently for different dyads. A study that logs only a session label loses the very structure that may explain relational meaning.

[@fig:dyadic_task_matrix] translates these distinctions into a staged study matrix. The matrix separates peer, mentor, and AI-assisted roles from co-located, remote, turn-taking, and persistent temporal-spatial conditions. It is a protocol-planning figure, not a claim that any one condition is universally superior.

Temporal Coordination: Turns, Overlap, and Persistence VI



PROTOCOL

generate_dyadic_task_matrix | Deterministic project-code figure; not participant outcome data unless stated

Temporal Coordination: Turns, Overlap, and Persistence

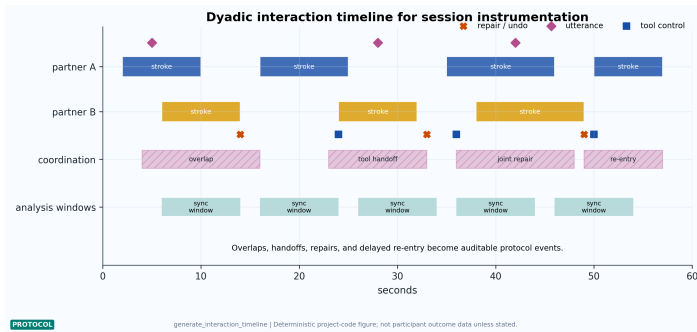
VII

Figure 1: Dyadic task matrix for staged DigiPPPiP studies. Values generated by `generate_dyadic_task_matrix()` encode conceptual mutual-responsiveness demand across role structures and temporal-spatial conditions; read rows as partner-role arrangements, columns as session modes, numeric labels as planning scores, and patterned cells as high-coordination conditions. The matrix supports the temporal architecture's sampling argument rather than ranking modes as universally better. Empirical calibration would require observed timing logs, participant reports, and matched-condition comparisons.

[@fig:interaction_timeline] shows how the temporal distinction becomes measurable. Strokes, utterances, repairs, and synchrony windows occupy separate tracks so future studies can distinguish overlap, alternation, latency, and delayed response without collapsing all of them into a single session label.

Temporal Coordination: Turns, Overlap, and Persistence

VIII



Temporal Coordination: Turns, Overlap, and Persistence IX

Figure 2: Instrumented dyadic timeline linking partner actions, awareness cues, repair events, replay windows, and analysis windows. The figure is generated by `generate_interaction_timeline()`; read labelled bars as partner strokes, patterned spans as coordination windows, markers as utterances, tool-control changes, and repairs, and bottom bands as candidate analysis intervals. It supports the claim that temporal mode must be logged rather than asserted. The plot is a protocol schematic; participant data would replace the simulated intervals.

[@fig:parallel_sequential_patterns] complements the timeline with three canonical event patterns. The point is design clarity: parallel marks, turn-taking marks, and long-delay marks invite different hypotheses about attention, anticipation, and narrative continuity.

Temporal Coordination: Turns, Overlap, and Persistence X

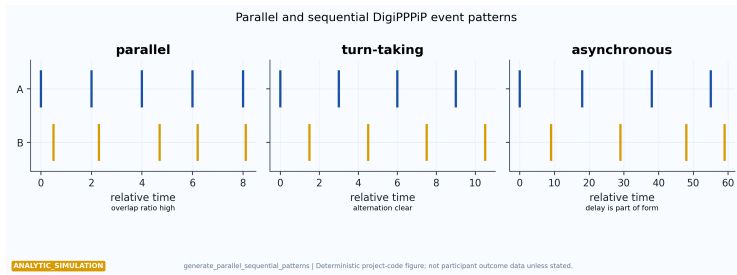
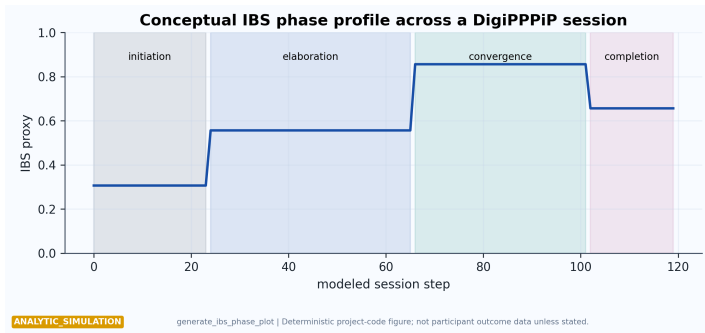


Figure 3: Event-pattern comparison for parallel, turn-taking, and asynchronous DigiPPiP sessions. The conceptual raster plots generated by `generate_parallel_sequential_patterns()` use blue and orange event ticks to show overlap ratio, alternation, and long-delay response structure. The figure supports the temporal taxonomy by showing why synchrony, alternation, and persistence are different hypotheses. It is schematic; empirical use would require timestamped session logs.

Temporal Coordination: Turns, Overlap, and Persistence XI

[@fig:ibs_phases] gives a conceptual phase model for synchronous sessions. It should not be read as empirical functional near-infrared spectroscopy (fNIRS) output. Its role is to make explicit the hypothesis that initiation, elaboration, convergence, and completion may differ in inter-brain-synchrony profiles and should be tested directly.



Temporal Coordination: Turns, Overlap, and Persistence

XII

Figure 4: Conceptual inter-brain-synchrony phase profile across initiation, elaboration, convergence, and completion. Values are generated by `generate_ibs_phase_plot()` from `src/hyperscanning.py`; read the line as an illustrative synchrony proxy and shaded bands as session phases. The figure supports hypothesis formation for synchronous sessions, not neural evidence. Real fNIRS or electroencephalography (EEG) analyses would need artifact correction, controls, and preregistered phase definitions.